

- 1 (a) Glycolysis is a biochemical pathway that occurs in the cytoplasm of cells.

In glycolysis, a molecule of glucose is metabolised to two molecules of pyruvate. The process is outlined in Fig. 1.1.

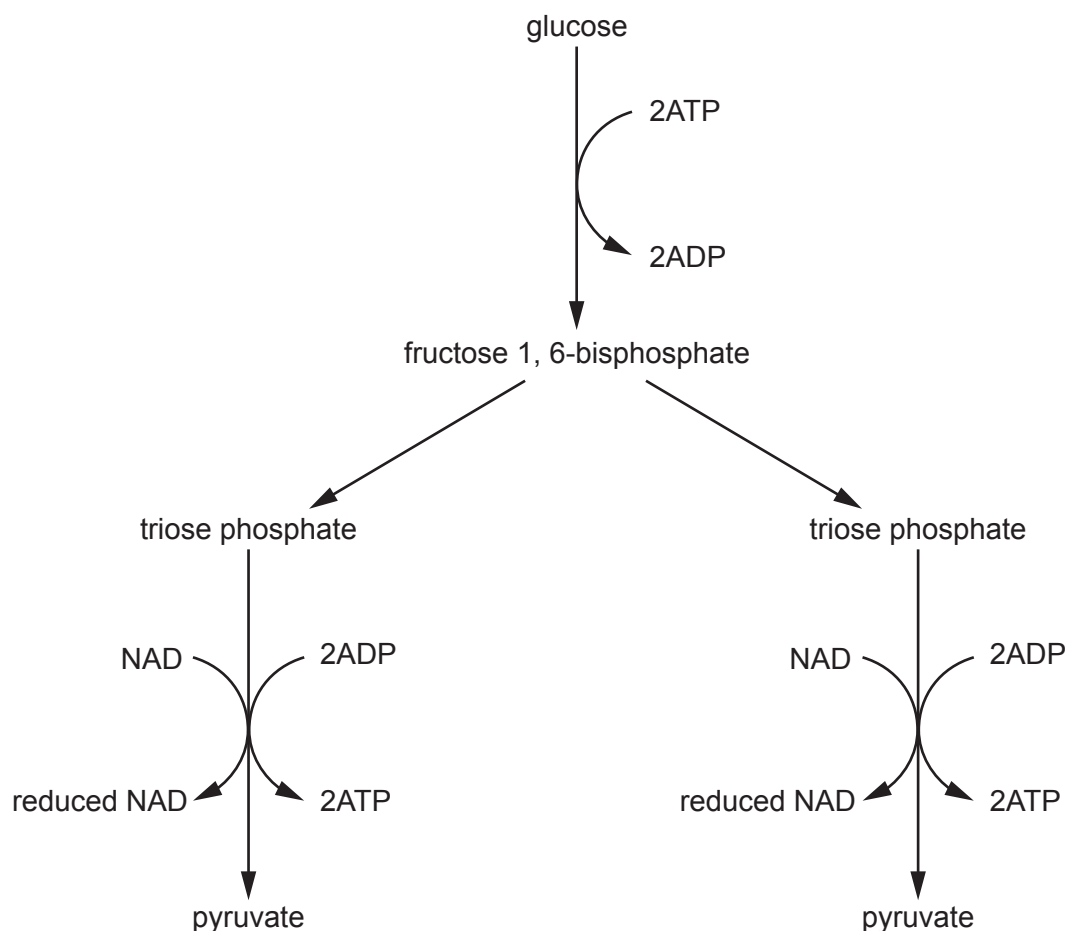


Fig. 1.1

- (i) Explain why glucose is phosphorylated at the beginning of glycolysis.

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..... [1]

- (ii) Suggest **one** use of the reduced NAD that is produced in glycolysis.

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..... [1]

- (iii) Name the type of phosphorylation reaction by which ATP is made during glycolysis.

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..... [1]

(b) Pyruvate can enter the mitochondrion by active transport.

Describe the main conditions that are required for pyruvate to enter the mitochondrion by active transport.

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..... [3]

(c) Pyruvate is involved in the link reaction in the matrix of the mitochondrion.

Describe the link reaction.

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..... [3]

[Total: 9]